

FITTRACK

COMBINED PROJECT DOCUMENTATION

Date: 29 September 2026

Project Detail	Details
Project Title	Fittrack
Team ID	SWTID-2026-9989
Team Leader	B Keerthana
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This document combines the uploaded Fittrack project templates into one project submission document. The structure follows the supplied templates and their terminology.

1. PROJECT INITIALIZATION AND PLANNING – DEFINE PROBLEM STATEMENTS

Date: 29 September 2026 | Team ID: SWTID-2026-9989 | Project Name: Fittrack

Customer Problem Statement

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A fitness beginner	Start and maintain a consistent fitness routine	I do not know which activities are suitable for my goals	There is limited personalized guidance and progress visibility	Confused and less motivated
PS-2	A regular fitness user	Track workouts and measure progress	My activities and goals are difficult to organize in one place	Fitness tracking is often separated from goal management	Frustrated and inconsistent
PS-3	A goal-oriented fitness user	Reach my fitness goals efficiently	I need recommendations based on my goals and activity history	Generic fitness advice may not match my needs	Unsure about the next step

Problem Statement Summary

Fittrack addresses the difficulty users face in maintaining fitness routines, recording activities, monitoring progress, and receiving useful goal-oriented guidance. The solution should make fitness management simple, organized, and personalized.

2. CUSTOMER JOURNEY MAP

Date: 29 September 2026 | Team ID: SWTID-2026-9989 | Project Name: Fittrack

Journey Stage	Stage 1: Discover & Register	Stage 2: Set Goals & Profile	Stage 3: Track Activities	Stage 4: View Progress	Stage 5: Get Recommendations
Actions	Find Fittrack; register	Enter fitness details; set goals	Record workout/activity	Review activity and goal progress	Request/view personalized guidance
Touchpoint	Registration/login screen	Fitness profile and goals	Workout/activity form	Progress dashboard	Recommendation feature
Customer Thought	Can this help me stay fit?	Are these goals right for me?	Is recording this easy?	Am I improving?	What should I do next?
Customer Feeling	Curious / hopeful	Motivated but uncertain	Engaged	Encouraged or concerned	Supported / confident
Process Ownership	Authentication module	Profile and goal module	Activity tracking module	Progress module	Recommendation module
Opportunity	Make registration simple	Provide clear goal options	Minimize data entry	Show understandable progress	Make recommendations relevant and actionable

The journey map follows the supplied template's focus on actions, touchpoints, customer thoughts, feelings, process ownership, and improvement opportunities.

3. SOLUTION REQUIREMENTS

Date: 29 September 2026 | Team ID: SWTID-2026-9989 | Project Name: Fittrack

Step 1: Functional Requirements

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Users shall securely register, log in, and verify their identity using authenticated sessions/JWT.	High
2	Authorization levels	The system shall control access to protected user information and administrative functions according to user roles.	High
3	External interfaces	The system may connect to an AI/recommendation service to provide personalized fitness guidance.	Medium
4	Transactions processing	The system shall validate, store, update, and retrieve fitness activities, goals, profiles, and progress data.	High
5	Reporting	The system shall provide progress information such as recorded activities, goal status, and fitness history.	High
6	Business rules, etc.	Activities shall belong to the authenticated user; goals shall be associated with the user; recommendations shall use available profile, goal, and activity information.	High
7	Compliance to laws or regulations	The system shall protect personal data, avoid unnecessary collection, and provide secure handling of user information. Specific legal certification is not	Medium

		claimed by the project.	
8	Other	Users shall be able to manage their fitness profile, record workouts, manage goals, and receive recommendations.	High

Step 2: Non-Functional Requirements

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Common API operations should respond quickly under normal development conditions.	Target: typical standard requests within 2 seconds; recommendation response may vary with external service.
2	Scalability	System should support increasing users and fitness records through modular backend and database design.	Should support growth without redesigning core modules.
3	Security & Data Privacy	Use authentication, authorization, input validation, secure secrets, and protected database access.	No passwords/secrets exposed in API responses or source code.
4	Reliability & Availability	Handle database/service failures gracefully and provide meaningful errors.	Failed requests return controlled error responses; no silent data corruption.
5	Usability & Accessibility	Fitness workflows should be simple, readable, and easy to navigate.	New user should be able to register, set a goal, and record an activity without technical assistance.
6	Maintainability & Portability	Use modular routes, controllers, models, and services.	Components can be updated independently; project can run on supported development environments.

4. DATA FLOW DIAGRAM (DFD)

Date: 29 September 2026 | Team ID: SWTID-2026-9989 | Project Name: Fittrack

The supplied DFD template defines external entities, numbered processes, data stores, and labeled data flows. The following Fittrack DFD specification can be used to draw the final diagram.

External Entities

Entity	Data Sent / Received
Fitness User	Registration details, login details, profile, goals, activity data; receives progress and recommendations.
Admin	User/platform management information; receives system status and management information.
Recommendation / AI Service	Receives relevant fitness context; returns recommendation content.

Processes

Process No.	Process	Input	Output
1.0	Authenticate User	Registration/login details	Authenticated user / JWT
2.0	Manage Fitness Profile	Profile details	Stored/updated profile
3.0	Manage Goals	Goal details	Stored/updated goals
4.0	Track Fitness Activity	Workout/activity details	Stored activity record
5.0	Calculate / Display Progress	Activities + goals	Progress information
6.0	Generate Recommendation	Profile + goals + activity context	Personalized recommendation

Data Stores

Data Store	Contents
D1 – User Database	User account and authentication-related information
D2 – Fitness Profile	Fitness level, preferences, and profile information
D3 – Goals Database	Fitness goals, targets, progress, and status
D4 – Activity Database	Workout/activity records and history
D5 – Recommendation Records	Generated/retrieved recommendation information, if persisted

Main Data Flows

- Fitness User → 1.0 Authenticate User: registration/login data.
- 1.0 Authenticate User → D1: validated user information.
- Fitness User → 2.0 Manage Fitness Profile: fitness profile and preferences.
- Fitness User → 3.0 Manage Goals: goal type and target.

- Fitness User → 4.0 Track Fitness Activity: activity type, duration, date, and notes.
- D3 + D4 → 5.0 Calculate / Display Progress: goals and activity history.
- D2 + D3 + D4 → 6.0 Generate Recommendation: profile, goals, and relevant activity context.
- 6.0 Generate Recommendation → Fitness User: personalized fitness recommendation.

5. TECHNOLOGY STACK

Date: 29 September 2026 | Team ID: SWTID-2026-9989 | Project Name: Fittrack

The supplied Technology Stack template asks for frontend/client, backend/server, database, deployment, version control/CI-CD, and third-party APIs/tools. The choices below are defined for Fittrack.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, JavaScript	Provides a simple responsive client interface for registration, profiles, goals, activity tracking, progress, and recommendations.
2	Backend / Server-Side	Node.js + Express.js	Supports REST API development and modular server-side application logic.
3	Database / Data Storage	MongoDB + Mongoose	Suitable for flexible fitness profile, activity, goal, and recommendation documents with schema validation through Mongoose.
4	Cloud / Hosting / Deployment	To be selected for final deployment	The supplied source does not specify a deployment provider; the choice should be finalized based on project hosting requirements.
5	Version Control & CI/CD	Git + GitHub	Supports team collaboration, source-code versioning, and project backup. CI/CD can be added if required.
6	Third-Party APIs / Other Tools	Postman / Thunder Client; AI/recommendation service if integrated	Postman/Thunder Client supports API testing. An AI/recommendation service can provide personalized fitness guidance.

Technology Stack Summary

- Frontend: HTML5, CSS3, JavaScript.
- Backend: Node.js with Express.js.
- Database: MongoDB with Mongoose.
- API Testing: Postman / Thunder Client.
- Version Control: Git and GitHub.
- Recommendation Layer: AI/recommendation service, if enabled in the implementation.

TEAM DETAILS AND SUBMISSION INFORMATION

Role	Name	Email
Team Leader	B Keerthana	Kamithakeerthana@gmail.com
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Note: The uploaded templates provide structures and prompts rather than Fittrack-specific implementation evidence. Where a technology or deployment detail was not specified in the uploaded source, it has been clearly marked as to be selected/if integrated rather than presented as an already completed fact.

Brainstorming & Idea Prioritization Template

Date	29 september 2026
Team ID	SWTID -2026-9989
Project Name	FitTrack
Maximum Marks	3 Marks

Step 1: Brainstorm and Idea Listing

Each team member lists out as many ideas as possible without judging them at this stage.

S.No	Team Member	Idea / Suggestion	Category	Group No.
1	Team Member 1	Secure User Authentication	Security	G1
2	Team Member 2	Workout CRUD Management	Functionality	G2
3	Team Member 3	Workout Search	Functionality	G2
4	Team Member 1	AI Workout Recommendations	AI/ML	G3
5	Team Member 2	AI Fitness Insights	AI/ML	G3
6	Team Member 3	Centralized Error Handling	Reliability	G1

Step 2: Idea Prioritization

Rate each grouped idea on feasibility and importance, then select the final idea(s) to move forward with.

Group No.	Final Idea	Feasibility (High/Medium/Low)	Importance (High/Medium/Low)	Priority	Selected (Yes/No)
1	Secure Authentication	High	High	1	Yes

2	Workout Management & Search	High	High	2	Yes
3	AI Recommendations & Fitness Insights	Medium	High	3	Yes

Code-Layout, Readability and Reusability

Date	15 March 2024
Team ID	XXXXXX
Project Name	XXXXXX
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code		
2	Proper File Structure	Files and folders are logically organized		
3	Meaningful Variable Names	Variables reflect their purpose clearly		
4	Function / Method Names	Functions are descriptively named		
5	Code Comments	Inline and block comments explain logic		
6	Modular Design	Code is split into reusable functions/modules		
7	No Redundant Code	Duplicate or unused code is removed		
8	Error Handling	Exceptions and errors are handled gracefully		

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1				

2				
3				
4				
5				

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure		
Readability		
Reusability		
Documentation / Comments		
Overall Score		

Solution Architecture

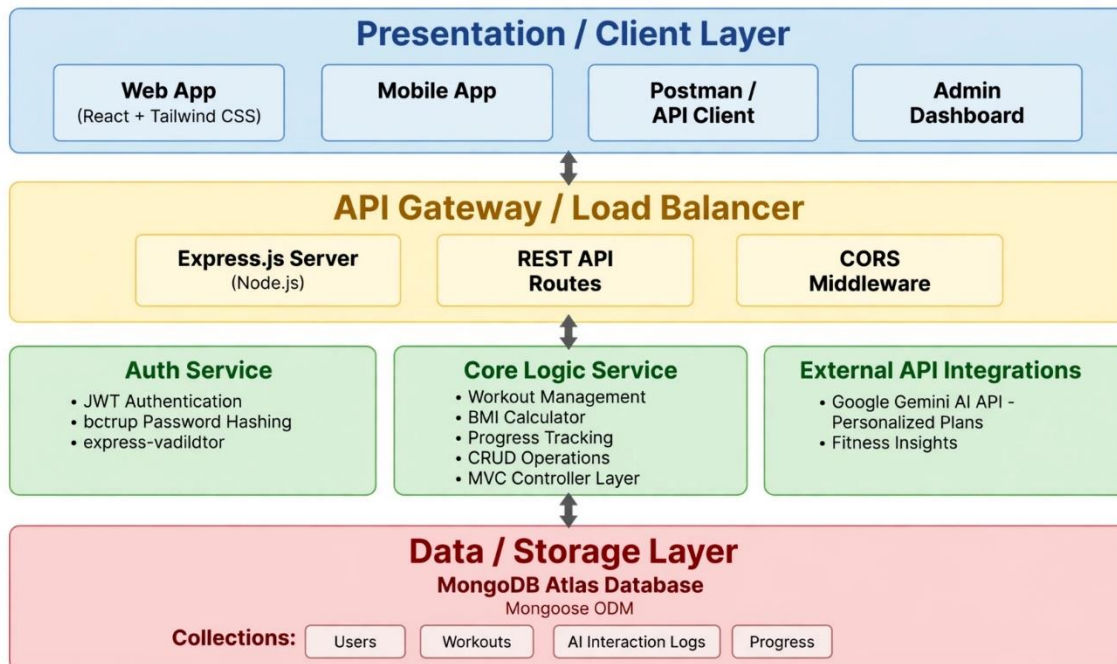
Date	29 september 2026
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Project Name	Fittrack
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.

Solution Architecture

Date: 15 March 2026 | Team ID: xxxxxx | Project Name: FitTrack - AI Powered Fitness Tracker | Maximum Marks: 5 Marks



FitTrack — AI Powered Fitness Tracker | Scalable • Secure • Real-Time Analytics

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
<i>Presentation Layer</i>	Handles user interface and client-side interaction. Users send requests and view JSON responses via UI.	React.js, Tailwind CSS, Postman, Mobile App
API Gateway/Load Balancer	Receives HTTP requests, routes to appropriate controllers, handles CORS error handling, validation	Express.js (Node.js), Express Router, CORS middleware, express-validator
Auth Service	Secures application, creates and verifies JWT tokens, hashes passwords	JWT, bcrypt, express-validator middleware
Core Logic/Business Logic	Processes business logic, workout CRUD, BMI calculation, progress tracking, request validation	MVC Pattern, Controllers, Services Layer, Mongoose Models

Initial Project Planning Template

Date	29 September 2026
Team ID	SWTID-2026-9989
Project Name	Fittrack
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create a product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members	Sprint Start Date	Sprint End Date (Planned)
Sprint-1	Registration	USN-1	As a user, I can register for the application by entering my email, and password, and confirming my password.	2	High	Member 1 Member 2	12 March 2026	13 March 2026
Sprint-1	Login	USN-2	As a user, I will receive a confirmation email once I have registered for the application	1	High	Member2 Member 4	14 March 2026	15 March 2026
Sprint-2	Workout add	USN-3	As a user, I can register for the application through Facebook	2	Low	Member4 Member 1	16. March 2026	17 March 2026

Sprint-1	Ai application	USN-4	As a user, I can register for the application Gemini ai through Gmail	2	Medium	Member2 Member 3	18 March 2026	19 March 2026
Sprint-1	Ai insights	USN-5	As a user, I can log into the application by entering email & password	1	High	Member1 Member 3	20 March 2026	21 March 2026



Problem-Solution Fit

Date	29 september 2026
Team ID	SWTID-2026-9989
Project Name	Fittrack
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] Fitness enthusiasts like Rahul, Gym goers, Beginners, People who workout daily at home/gym, Personal trainers. Age 18-45, regular exercise pandravanga but digital tracking venum nu nenaikravanga.	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] Budget low, cannot afford personal trainer monthly. Device: Only smartphone / low-end laptop irukku (8GB RAM, 1GB storage, Intel i5 / Ryzen 5). No technical knowledge, simple UI than venum. Internet limited, so lightweight app venum.	5. AVAILABLE SOLUTIONS PROS & CONS [AS] - *Notebooks / Spreadsheets:* Pros – Easy, No internet needed. Cons – No security, No search, No AI, Data loss easy, Time-consuming. - *Existing Apps (MyFitnessPal, Nike Training):* Pros – Tracking irukku. Cons – Costly subscription, Complex UI, No personalized AI plan
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		with Age/Goal/Experience, Data privacy illa. - *Personal Trainer:* Pros – Expert guidance. Cons – Very costly, Not available 24x7.
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] - Difficulty maintaining centralized workout record (Daily) - Manual tracking in notebook/spreadsheet leads to inaccurate history (Daily) - Unable to search previous workouts efficiently (Weekly) - No personalized workout recommendations (Every workout time) - No insights on total workouts, avg duration, calories (Weekly) - Time-consuming manual calculation of calories & duration (Daily) - No secure user authentication for personal fitness records	9. PROBLEM ROOT / CAUSE [RC] Root cause: Centralized, secure, intelligent system illa. Manual process than main problem. Data validation illa, JWT auth illa, AI integration illa, Search & reporting system illa. MVC architecture maathiri proper structure illa. Google Gemini maathiri AI use pannala.	7. BEHAVIOR + ITS INTENSITY [BE] - Daily manual la workout note pannuvanga – High intensity - Google la “best workout for weight loss” nu search pannuvanga – Medium intensity - YouTube videos paathu workout pannuvanga but consistency illa – High intensity - Spreadsheets la calories calculate pannuvanga – Low intensity, but frustrate aayidu Anga
3. TRIGGERS TO ACT [TR] - Gym join pannum bothu progress track pannanum nu thonuthu - Weight loss / muscle gain goal set pannum bothu - Notebook la data tholaichathuna	10. YOUR SOLUTION [SL] *MuscleUp – AI Powered Fitness Ecosystem* Backend lo http://Node.js + Express + MongoDB MVC architecture, JWT Authentication, bcrypt	8. CHANNELS OF BEHAVIOR [CH] *ONLINE:* Google Search, YouTube Fitness Channels, Instagram Fitness Pages, Mobile Apps, Postman / Web App use pannuvanga.

- Friend vera app use pannu progress kaamikkum bothu - Doctor fitness report ketta bothu	password hashing, express-validator. Frontend lo React + Tailwind simple UI, 8GB RAM / 1GB storage device la kooda smooth ah work aagum.	
4. EMOTION BEFORE / AFTER [EM]* *Before:* Frustrated, Confused, Demotivated, No clarity if exercise correct ah nu, Insecure about fitness level. *After:* Confident, Motivated, Organized, Clear progress theriyum, Professional guidance irukku n BEFORE / AFTER [EM]	Core Feature: *Gemini AI Integration* - Age, Goal, Experience vachu personalized Workout & Diet plan generate pannum. Search & Filter exercises, BMI Calculator, Progress Tracking, Reporting Dashboard ellam irukku. Low cost, 24x7 available personal trainer maathiri work pannum.	OFFLINE: * Gym, Home workout, Notebook la writing, Friends kitta advice kekkarathu.

Project Initialization and Planning Phase

Date	29 September 2026
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Project Title	Fittrack
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	To revolutionize fitness journey by implementing AI-driven personalized workout & diet plans, secure JWT authentication, and centralized progress tracking.
Scope	The project comprehensively assesses fitness tracking, incorporates Node.js MVC, React frontend, and Gemini AI for a robust and efficient system. Supports up to Intel i5/Ryzen 5, 8GB RAM, 1GB Storage.
Problem Statement	
Description	Addressing inaccuracies in manual tracking, lack of personalized guidance, no security in spreadsheets, and high cost of personal trainers.

Impact	. Solving will improve user consistency, reduce frustration, provide data security with bcrypt + JWT, and make fitness affordable 24x7.
Proposed Solution	
Approach	Employing MERN Stack + Google Gemini AI to generate dynamic workout/diet plans based on Age, Goal, Experience. MVC architecture, express-validator, search & filter.
Key Features	Gemini AI personalized plan generation – JWT Secure Auth & Bcrypt – Workout Management with Search – BMI Calculator & Progress Dashboard Reporting & Analytics



	- Real-time decision-making for quicker loan approvals. - Continuous learning to adapt to evolving financial landscapes.
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel i5 / Ryzen 5, 4 cores minimum
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	1 GB minimum / 256GB SSD recommended
Software		

Frameworks	MERN frameworks	Node.js, Express.js, React.js, mongodb
Libraries	Additional libraries	Mongoose, JWT, bccrpt, express-valalitor, Tailwind CSS, Google Gemini API
Development Environment	IDE	VS Code, Postman, MongoDB Atlas, GitHub
Data		
Data	Source, size, format	User fitness profiles, ExerciseWorkout logs – JSON format from MongoDB dataset (1000+ exercises)

Technology Stack

Date	29 september 2026
Team ID	SWTID-2026-9989
Project Name	Fittrack
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	React.js, HTML5, CSS3, Axios, React-Router-DOM	Component -based reusable UI, responsive design, fast virtual DOM rendering, easy state management. Axios for API calls.

architecture and provide a brief justification for your choice.

Technology Stack Details

2	Backend / Server-Side	Node.js (v16+), Express.js	Fast, non-blocking, lightweight routing for RESTful APIs. Same JavaScript stack as frontend, easy to handle 50+ concurrent users. MVC pattern for maintainability.
3	Database / Data Storage	MongoDB Atlas, Mongoose ODM	NoSQL flexible schema for workout data, handles unstructured data easily, BSON document storage. Mongoose for schema validation, indexing on user field for fast search.
4	Cloud / Hosting / Deployment	Render /Vercel (Backend), MongoDB Atlas Cloud, .env for config	Cost-effective hosting, easy scalability, auto-deployment from GitHub, 99.9% uptime..env for secure secret management.

5	Version Control & CI/CD	Git, GitHub	Seamless n, version history, easy rollback, automated deployment via GitHub integration.
6	Third-Party APIs / Other Tools	Google Gemini AI API, JWT, bcrypt.js, Postman, VS Code	Gemini API Personalize d workout plan & fitness insights (Age, Goal, Experience input). JWT bcrypt.js (salt 10) -> Password hashing. Postman -> API testing & documenta tion.